CRIMEVISIONAI

Scalability & Future Plan

Folder 8 | Current limits and staged roadmap | Prepared 23 October 2026

Project: CrimeVisionAI — Multimodal Forensic Evidence Analysis and Investigation Support System

Document status: Project-specific draft based on the implementation information currently available. Verify placeholders and run-dependent results before formal submission.

1. Current limitations

Limitation	Impact	Priority
Local inference and model initialization	Latency and memory use vary by hardware; concurrent analysis may block requests.	High
SQLite + local filesystem	Limited horizontal scaling, concurrency and shared storage capabilities.	Medium for prototype; high before multi-user deployment
100 MB upload cap and video cap	Large files may be rejected or truncated by configured processing limits.	Medium
Video sampling at 1 FPS	Short events between sampled frames can be missed.	High for time-sensitive events
Q&A grounding gap	Answers may omit relevant detected entities or provide incomplete synthesis.	High
No speaker diarization	Cannot reliably assign transcript segments to speakers.	Medium
Security / chain-of-custody controls unverified	Unsuitable to claim secure production or evidentiary integrity.	High before real-case use
Performance benchmarks missing	No defensible capacity or latency claim.	High before public claims

2. Scalability plan

Aspect	Current state	Proposed upgrade
User load	Local Flask development server	Production WSGI server, reverse proxy, request timeouts and measured concurrency.
Background jobs	Synchronous / current behavior must be verified	Queue long-running analysis in a worker system with progress/status records.
Data storage	SQLite + local filesystem	PostgreSQL for multi-user metadata and object storage for evidence, with access controls.
Model serving	Local CPU / auto CUDA detection	Dedicated inference workers, model caching, resource quotas and batch scheduling.
Video processing	1 FPS, max 180 seconds / 180 frames	Configurable sampling and scene-change selection after accuracy/performance evaluation.
Observability	Not confirmed	Structured logs, metrics, request IDs, model timing and safe error reporting.
Security	Local development configuration	Authentication/authorization, HTTPS, secrets management, audit trail and retention policies.

3. Future roadmap

Phase	Enhancement	Expected impact / exit criteria
Phase 1	Fix Q&A retrieval/grounding and entity coverage	Regression tests cover known omissions; answers link to source evidence.
Phase 2	Timeline integrity and cross-modal event fusion	Consistent timestamps, provenance, duplicate handling and conflict surfacing.
Phase 3	PDF report export and audit history	Reproducible report with case metadata, source links, model caveats and review status.
Phase 4	Repeatable model and performance evaluation	Documented test dataset, metrics, latency and hardware profile.
Phase 5	Secure multi-user deployment	Access controls, secure file storage, disabled debug mode and monitored deployment.
Phase 6	Speaker diarization / advanced video analysis / 3D visualization	Implement only after requirements, feasibility and validation criteria are defined.

4. Scalability conclusion

Scale should follow correctness and governance, not precede them. First make outputs traceable and Q&A; grounded; then establish performance baselines; then replace prototype storage/runtime components where measurements justify the change. A scalable architecture alone does not establish forensic validity.